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NORMAL GROWTH

khanirfan0392@gmail.com

Phases of Growth

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		Time period
Prenatal period :		
a. Zygote/fertilized ovum.		< 2 wks
b. Embryo.		3 - 8 wks
c. Fetus.		> 9 wks
Perinatal period		22 wks of gestation upto 7 days after birth
Postnatal growth	a. Newborn	1 st 4 wks { 0 - 7 days : Early 7 - 28 days : Late
	b. Infancy	1 st year
	c. Toddler	1 - 3 yrs
Preschool child		3 - 6 yrs
School age		6 - 12 yrs
Adolescence		10-19 yrs { Early : 10 - 13 yrs Mid : 14 - 16 yr Late : 17 - 19 yrs

PUBERTAL CHANGES :

	Females	Males
Age of onset	8 - 13 yrs	9 - 14 yrs
Sequence	Thelarche (Development of breast) ↓ Pubarche ↓ Growth spurt (↑ height) ↓ Menarche	↑ size of testis : Assess with Prader's orchidometer ↓ ↑ size of penis ↓ Pubarche { Voice changes Pubic hair appearance Spermarche ↓ Growth spurt ↓ Axillary & facial hair

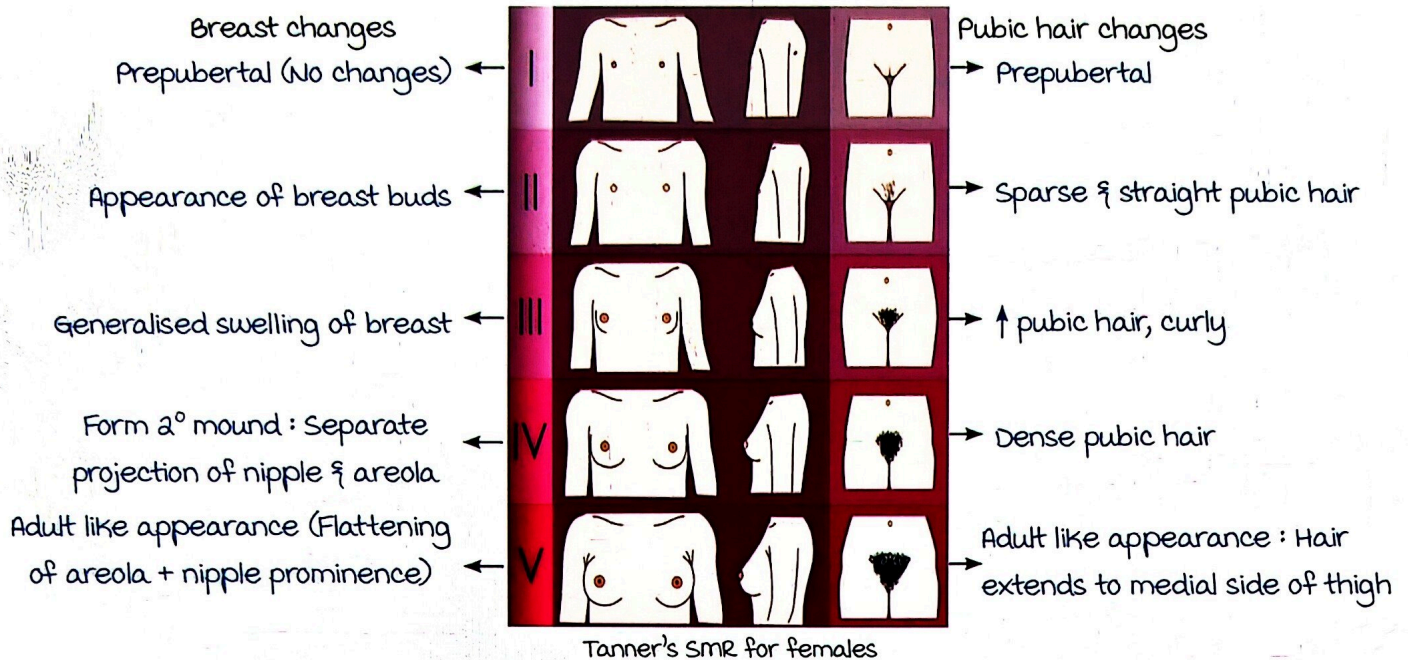
Note : Prader's orchidometer → Contains beads that correspond to size of testis.

Tanner's sexual maturity rating (SMR) :

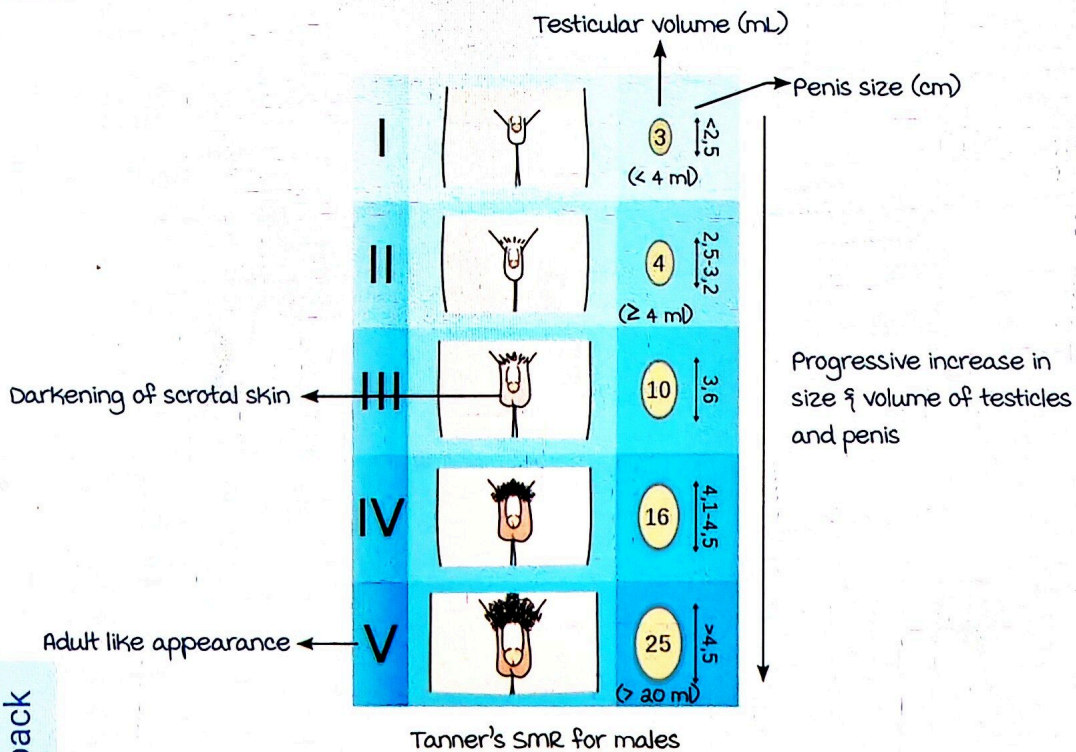
used to assess puberty : Based on external appearance.

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Females :



males :



Note : Pubic hair changes are the same in males and females.

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Laws of Growth

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PATTERN OF GROWTH :**1. Cephalocaudal (Head to toe) :**

Growth of head → Growth of the lower body and limbs.
(in utero) (After birth)

2. Limbs :

- Distal growth → Proximal growth.
Eg : Fingers will grow first, then the arms.

3. Different tissues grow at different rates :

Eg : Brain tissue : 90 % of growth is completed in **2 years**.

Gonadal growth : ↑ around puberty → Production of sex steroids.

Lymphoid growth : Tonsils, lymph nodes (Become palpable).

- Rapid during 4-8 yrs of age.

Skeletal growth : Steady process until puberty

↓
Growth spurt at puberty
↓ Post puberty
Completion/maturation : Fusion of epiphysis { **18 yrs** in females
20-22 yrs in males
(Generally taller)

Anthropometry

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methods to assess growth.

WEIGHT :

- Average birth weight in India : **2.8 - 3 kg**.
- < 2.5 kg → Low birth weight (LBW).

Normal weight loss after birth :

- D/t extracellular fluid loss.
 - 10 % of body weight.
- } 1st week after birth.

Time after birth	Weight gain/loss
upto days	↓ by 10 % (14 % in preterm)
By 10 days (By 14 days in preterm)	Regains birth weight
upto 3 months	↑ 20 - 40 g/day
upto 1 year	↑ 400 g/month
upto 7 years	↑ 2 kg/year
After 7 yrs	↑ 3 kg/year

With respect to weight :

- Doubles : 5 - 6 months.
- Triples : 1 year.
- Quadruples : 2 years.

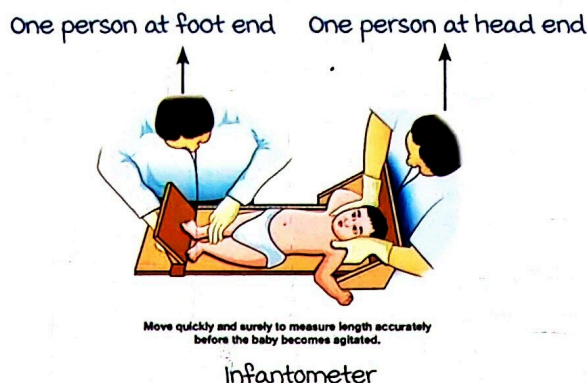
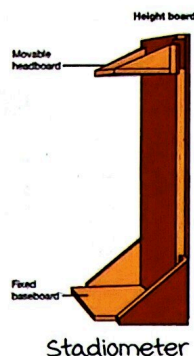
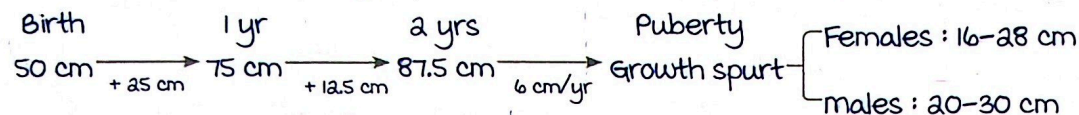
Normal Growth

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LENGTH/HEIGHT :

- Length : upto 2 yrs old → measured while supine with infantometer.
- Height : > 2 yrs old → measured while standing with stadiometer.

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Equipments :**Pattern of increase in height :****Arm span :**

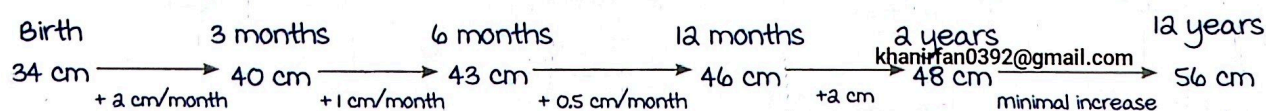
In un co-operative children :

Distance b/w middle fingers
of outstretched hands.

- Birth : Height > arm span by 2.5 cm.
- At 11 yrs of age : Height = Arm span.
- > 11 yrs of age : Height < arm span by 1-2 cm.

CIRCUMFERENCES :**Head circumference (HC) :** Reflection of brain growth.

Pattern of ↑ in HC :

**Chest circumference (CC) :**measured at 4th intercostal space (roughly at the level of nipples) in children.

HC v/s CC :

At birth : HC > CC (By 3 cm).

By 1 year : HC = CC.

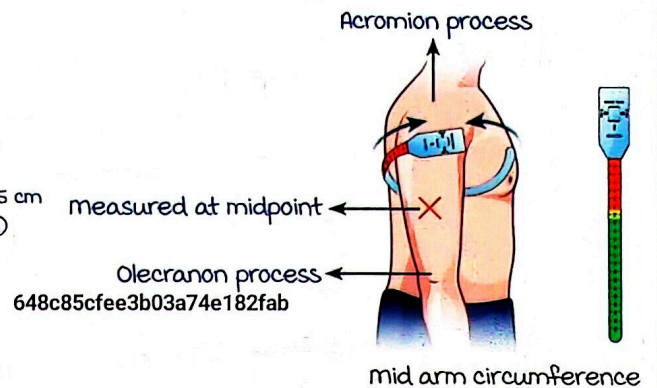
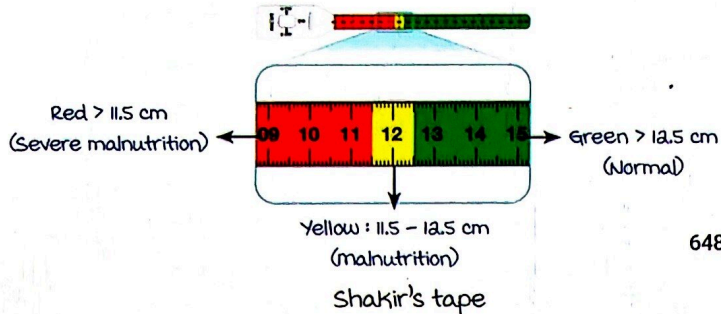
After 1 year : CC > HC.

Feedback

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mid arm circumference (MAC) :

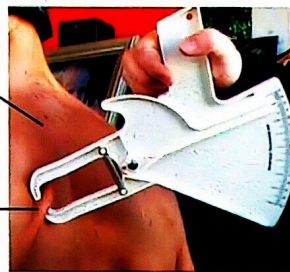
- measured in age groups of 1-5 yrs.
- Depends on **muscle mass & fat** → Indicator of nutritional status.
- measured at community level :

**SKIN FOLD THICKNESS :**

Sites :
Biceps, subscapular, triceps

Assessment of
subcutaneous fat

Nutritional status



Harpenden's calliper

Normal : ≥ 10 mm

Severe malnutrition : < 6 mm

Growth Monitoring

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monitored by plotting on growth charts.

Types :

1. WHO charts : For < 5 yrs.
2. Indian Academy of Pediatrics (IAP) chart : 5-18 yrs.
3. Consolidated charts.

WHO CHART :

- Since 2006.
- Internationally used.

Chart based on **multicenteric growth reference study (MGRS)** with :

Children

Who :

- Had **no developmental anomalies**.
- Exclusively **breastfed for 6 months**.

From :

Developed & underdeveloped countries

- India.
- USA.
- Oman.
- Brazil.
- Ghana.
- Norway.

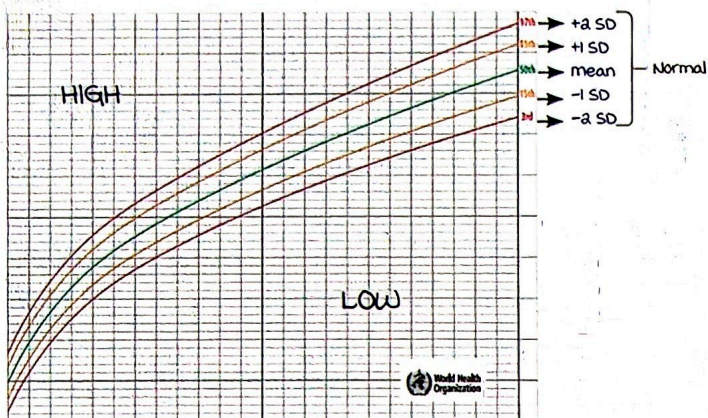
Colour of WHO chart :

- Blue (Boys).
- Pink (Girls).

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Variants :

1. Percentile : Frequency distribution of a given value.
2. Standard deviation (SD) : Deviation from mean value (50th percentile).



Standard deviation & percentiles in growth chart

Interpretation of WHO growth chart :

Stature	Height for age
Stunting/ short stature	< 3 rd percentile or < -2 SD
Severe stunting	< -3 SD
Undernutrition	Weight for height
Wasting	< 3 rd percentile / < -2 SD
Severe wasting	< -3 SD
Status	Weight for age
Underweight	< 3 rd percentile / < -2 SD
Severely underweight	< -3 SD
Status	Weight for height
Overweight	+ 2 to + 3 SD
Obesity	> + 3 SD

IAP CHARTS :

1. Height for age chart : Assess stunting : < -2 SD or < 3rd percentile.
2. BMI chart :

